ACVP KAS FFC JSON Specification

Barry Fussell
Cisco Systems, Inc.
170 West Tasman Drive, San Jose, California

Russell Hammett
G2, Inc.
302 Sentinel Drive, Suite #300, Annapolis Junction, MD 20701

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Abstract

This document defines the JSON schema for testing SP800-56a KAS FFC implementations with the ACVP specification.

Keywords

The following are keywords to be used by search engines and document catalogues.

ACVP; cryptography

Foreword

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Audience

This document is intended for the users and developers of ACVP.

Conventions

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 of [RFC 2119] and [RFC 8174] when, and only when, they appear in all capitals, as shown here.

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Executive Summary

The Automated Crypto Validation Protocol (ACVP) defines a mechanism to automatically verify the cryptographic implementation of a software or hardware crypto module. The ACVP specification defines how a crypto module communicates with an ACVP server, including crypto

capabilities negotiation, session management, authentication, vector processing and more. The ACVP specification does not define algorithm specific JSON constructs for performing the crypto validation. A series of ACVP sub-specifications define the constructs for testing individual crypto algorithms. Each sub-specification addresses a specific class of crypto algorithms. This sub-specification defines the JSON constructs for testing SP800-56a KAS FFC implementations using ACVP.

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1. Introduction

The Automated Crypto Validation Protocol (ACVP) defines a mechanism to automatically verify the cryptographic implementation of a software or hardware crypto module. The ACVP specification defines how a crypto module communicates with an ACVP server, including crypto capabilities negotiation, session management, authentication, vector processing and more. The ACVP specification does not define algorithm specific JSON constructs for performing the crypto validation. A series of ACVP sub-specifications define the constructs for testing individual crypto algorithms. Each sub-specification addresses a specific class of crypto algorithms. This sub-specification defines the JSON constructs for testing SP800-56a KAS FFC implementations using ACVP.

2. Supported KAS-FFCs

The following key derivation functions **MAY** be advertised by the ACVP compliant cryptographic module:

- KAS-FFC / null / 1.0
- KAS-FFC / Component / 1.0

3. Test Types and Test Coverage

The ACVP server performs a set of tests on the KAS protocol in order to assess the correctness and robustness of the implementation. A typical ACVP validation session **SHALL** require multiple tests to be performed for every supported permutation of KAS capabilities. This section describes the design of the tests used to validate implementations of KAS algorithms.

3.1. Test Types

There are two test types for KAS testing:

- "AFT" — Algorithm Function Test. In the AFT test mode, the IUT **SHALL** act as a party in the Key Agreement with the ACVP server. The server **SHALL** generate and provide all necessary information for the IUT to perform a successful key agreement; both the server and IUT **MAY** act as party U/V, as well as recipient/provider to key confirmation.
- "VAL" — Validation Test. In the VAL test mode, The ACVP server **MUST** generate a complete (from both party U and party V's perspectives) key agreement, and expects the IUT to be able to determine if that agreement is valid. Various types of errors MSUT be introduced in varying portions of the key agreement process (changed DKM, changed key, changed hash digest, etc), that the IUT **MUST** be able to detect and report on.

3.2. Test Coverage

The tests described in this document have the intention of ensuring an implementation is conformant to [SP 800-56A Rev. 2].

3.2.1. KAS-FFC Requirements Covered

- SP 800-56Ar2 — 4.1 Key Establishment Preparations. The ACVP server is responsible for generating domain parameters as per the IUT's capability registration.
- SP 800-56Ar2 — 4.2 Key-Agreement Process. Both the ACVP server and IUT participate in the Key Agreement process. The server and IUT can both take the roles of party U/V, and as such the "performer" of steps depicted in "Figure 2: Key Agreement process" can vary.
- SP 800-56Ar2 — 5.1 Cryptographic Hash Functions. All modes of performing KAS **SHALL** make use of a hash function. The hash function **MAY** be used for validation of a successfully generated shared secret Z *noKdfNoKc), or as a primitive within the KDF being tested (kdfNoKc and kdfKc).
- SP 800-56Ar2 — 5.2 Message Authentication Code (MAC) Algorithm. A MAC is utilized for confirmation of success for kdfNoKc and kdfKc modes of KAS. Note — a MAC prerequisite is **REQUIRED** only for kdfKc, though is utilized for both kdfNoKc and kdfKc.
- SP 800-56Ar2 — 5.4 Nonce. Nonces are made use of in various KAS schemes — both the ACVP server and IUT **SHALL** be expected to generate nonces.
- SP 800-56Ar2 — 5.5 Domain Parameters. Domain Parameter Generation **SHALL** be performed solely from the ACVP server, with constraints from the IUTs capabilities

registration. The same set of domain parameters **SHALL** generate all keypairs (party U/V, static/ephemeral) for a single test case.

- SP 800-56Ar2 — 5.6 Key-Pair Generation. While Key-Pairs are used in each KAS scheme, the generation of said key-pairs is out of scope for KAS testing. Random tests from the VAL groups, **MAY** inject bad keypairs that the IUT **MUST** be able detect. These random tests are only present in groups given appropriate assurance functions see: Section 4.4
- SP 800-56Ar2 — 4.3 DLC-based Key-Transport Process / 5.7 DLC Primitives. Depending on the scheme used, either Diffie Hellman or MQV **SHALL** be used to negotiate a shared secret of z. Testing and validation of such key exchanges is covered under their respective schemes.
- SP 800-56Ar2 — 5.8 Key-Derivation Methods for Key-Agreement Schemes. All schemes/ modes save noKdfNoKc (component) **MUST** make use of a KDF. KDF construction **SHALL** utilize Section 4.10.1 for its pattern.
- SP 800-56Ar2 — 5.9 Key Confirmation. Most KAS schemes allow for a Key Confirmation process, the ACVP server and IUT **MAY** be Providers or Recipients of said confirmation. Additionally, key confirmation **MAY** be performed on one or both parties (depending on scheme).
- SP 800-56Ar2 — 6 Key Agreement Schemes. All schemes specified in referenced document are supported for validation with the ACVP server.

3.2.2. KAS-FFC Requirements Not Covered

- SP 800-56Ar2 — 4.1 Key Establishment Preparations. The ACVP server **SHALL NOT** make a distinction between IUT generated keys via a trusted third party and the IUT itself.
- SP 800-56Ar2 — 5.3 Random Number Generation. The IUT **MUST** perform all random number generation with a validated random number generator. A DRBG is **REQUIRED** as a prerequisite to KAS, but **SHALL NOT** be in the scope testing assurances.
- SP 800-56Ar2 — 5.4 Nonce. Nonce generation is utilized for several schemes. The various methods of generating a nonce described in section 5.4 **MUST** be used, however their generation **SHALL NOT** be in scope of KAS testing assurances.
- SP 800-56Ar2 — 5.5.2 Assurances of Domain-Parameter Validity. The ACVP server **SHALL** generate all domain parameters, IUT validation of such parameters is **SHALL NOT** be in scope for KAS testing.
- SP 800-56Ar2 — 5.5.3 Domain Parameter Management. Domain Parameter Management **SHALL NOT** be in scope for KAS testing.
- SP 800-56Ar2 — 5.6 Key-Pair Generation. While Key-Pairs **MUST** be used in each KAS scheme, the generation, assurances, and management of said key-pairs **SHALL NOT** be in scope of KAS testing.
- SP 800-56Ar2 — 5.8 Key-Derivation Methods for Key-Agreement Schemes. Two-step Key-Derivation (Extraction-then-Expansion) **SHALL NOT** be utilized in KAS testing.

- SP 800-56Ar2 — 5.7 Rationale for Selecting a Specific Scheme. It is expected that the IUT registers all schemes it supports in its capabilities registration. Selecting specific schemes from a KAS testing perspective **SHALL NOT** be in scope.
- SP 800-56Ar2 — 8 Key Recovery. Key Recovery **SHALL NOT** be in scope of KAS testing.

4. Capabilities Registration

ACVP requires crypto modules to register their capabilities. This allows the crypto module to advertise support for specific algorithms, notifying the ACVP server which algorithms need test vectors generated for the validation process. This section describes the constructs for advertising support of KAS FFC algorithms to the ACVP server.

The algorithm capabilities **MUST** be advertised as JSON objects within the 'algorithms' value of the ACVP registration message. The 'algorithms' value is an array, where each array element is an individual JSON object defined in this section. The 'algorithms' value is part of the 'capability_exchange' element of the ACVP JSON registration message. See the ACVP specification [ACVP] for more details on the registration message.

4.1. Prerequisites

Each algorithm implementation **MAY** rely on other cryptographic primitives. For example, RSA Signature algorithms depend on an underlying hash function. Each of these underlying algorithm primitives must be validated, either separately or as part of the same submission. ACVP provides a mechanism for specifying the required prerequisites:

Prerequisites, if applicable, **MUST** be submitted in the registration as the `prereqVals` JSON property array inside each element of the `algorithms` array. Each element in the `prereqVals` array **MUST** contain the following properties

Table 1 — Prerequisite Properties

JSON Property	Description	JSON Type
algorithm	a prerequisite algorithm	string
valValue	algorithm validation number	string

A "valValue" of "same" **SHALL** be used to indicate that the prerequisite is being met by a different algorithm in the capability exchange in the same registration.

An example description of prerequisites within a single algorithm capability exchange looks like this

```
"prereqVals":
[
  {
    "algorithm": "Alg1",
    "valValue": "Val-1234"
  },
  {
    "algorithm": "Alg2",
    "valValue": "same"
  }
```

]

Figure 1

4.2. Required Prerequisite Algorithms for KAS FFC Validations

Some algorithm implementations rely on other cryptographic primitives. For example, IKEv2 uses an underlying SHA algorithm. Each of these underlying algorithm primitives must be validated, either separately or as part of the same submission. ACVP provides a mechanism for specifying the required prerequisites:

Table 2 — Required Prerequisite Algorithms JSON Values

JSON Value	Description	JSON type	Valid Values	Optional
algorithm	a prerequisite algorithm	value	CCM, CMAC, DRBG, DSA, HMAC, SHA	No
valValue	algorithm validation number	value	actual number or “same”	No
prereqAlgVal	prerequistie algorithm validation	object with algorithm and valValue properties	see above	Yes

KAS has conditional prerequisite algorithms, depending on the capabilities registered:

Table 3 — Prerequisite requirement conditions

Prerequisite Algorithm	Condition
DRBG	Always **REQUIRED**
SHA	Always **REQUIRED**
DSA	DSA.PQGGen validation **REQUIRED** when IUT using assurance function of “dpGen”. DSA.PQGVer validation **REQUIRED** when IUT using assurance function of “dpVal”. DSA.KeyPairGen validation **REQUIRED** when IUT using assurances functions of “keyPairGen”, or “keyRegen”.
AES-CCM	AES-CCM validation **REQUIRED** when IUT is performing KeyConfirmation (KC) and utilizing AES-CCM.
CMAC	CMAC validation **REQUIRED** when IUT is performing KeyConfirmation (KC) and utilizing CMAC.

Prerequisite Algorithm	Condition
HMAC	HMAC validation **REQUIRED** when IUT is performing KeyConfirmation (KC) and utilizing HMAC.

4.3. KAS FFC Algorithm Capabilities JSON Values

Each algorithm capability advertised is a self-contained JSON object using the following values.

Table 4 — KAS FFC Capabilities JSON Values

JSON Value	Description	JSON type	Valid Values	Optional
algorithm	The algorithm under test	value	KAS-FFC	No
mode	The algorithm mode.	value	Component	Yes
revision	The algorithm testing revision to use.	value	“1.0”	No
prereqVals	Prerequisite algorithm validations	array of prereqAlgVal objects	See Section 4.2	No
function	Type of function supported	array	See Section 4.4	No
scheme	Array of supported key agreement schemes each having their own capabilities	object	See Section 4.5.1	No

Note: Some optional values are **REQUIRED** depending on the algorithm. Failure to provide these values will result in the ACVP server returning an error to the ACVP client during registration.

4.4. Supported KAS FFC Functions

The following function types **MAY** be advertised by the ACVP compliant crypto module:

- dpGen — IUT can perform domain parameter generation (FFC only)
- dpVal — IUT can perform domain parameter validation (FFC only)
- keyPairGen — IUT can perform keypair generation.
- fullVal — IUT can perform full public key validation ([SP 800-56A Rev. 2] section 5.6.2.3.1 / 5.6.2.3.3)
- ACVP server **MAY** inject keys into “VAL” type tests that will fail full public key validation.

- keyRegen — IUT can regenerate keys given a specific seed and domain parameter (pqg for FFC, curve for ECC)

4.5. KAS FFC Schemes

4.5.1. KAS FFC Scheme Capabilities JSON Values

All other scheme capabilities are advertised is a self-contained JSON object using the following values. Note that at least one of "noKdfNoKc", "kdfNoKc", or "kdfKc" **MUST** be supplied with the registration. See Section 4.5.2 for allowed FFC scheme types.

Table 5 — KAS FFC Capabilities JSON Values

JSON Value	Description	JSON type	Valid Values	Optional
kasRole	Roles supported for key agreement	array	initiator and/or responder	No
noKdfNoKc	Indicates no KDF, no KC tests are to be generated. Note this is a COMPONENT mode only test. This property **MUST** only be used with "KAS-FFC" / "Component"	object	Section 4.6.1	Yes
kdfNoKc	Indicates KDF, no KC tests are to be generated. Note this is a KAS-FFC only test. This mode **MUST** only be used for registrations with "KAS-FFC" (no mode)	object	Section 4.6.2	Yes
kdfKc	Indicates KDF, KC tests are to be generated. Note this is a KAS-FFC only test. This mode **MAY** only be used for registrations with "KAS-FFC" (no mode)	object	Section 4.6.3	Yes

4.5.2. Supported KAS FFC Schemes

The following schemes **MAY** be advertised by the ACVP compliant crypto module:

- dhHybrid1
- MQV2
- dhEphem — KeyConfirmation not supported.
- dhHybridOneFlow
- MQV1
- dhOneFlow — Can only provide unilateral key confirmation party V to party U.

- dhStatic

4.6. KAS FFC Modes

4.6.1. KAS FFC noKdfNoKc

Contains properties **REQUIRED** for "noKdfNoKc" registration.

Table 6 — NoKdfNoKc Capabilities

JSON Value	Description	JSON type	Valid Values	Optional
parameterSet	The parameter sets supported	object	Section 4.7.1	No

4.6.2. KAS FFC kdfNoKc

Contains properties **REQUIRED** for "kdfNoKc" registration.

Table 7 — kdfNoKc Capabilities

JSON Value	Description	JSON type	Valid Values	Optional
kdfOption	The kdf options supported	object	Section 4.10	No
dkmNonceTypes	The dkmNonceTypes supported	array of string	randomNonce, timestamp, sequence, timestampSequence	Required for dhStatic scheme
parameterSet	The parameter sets supported	object	Section 4.7.1	No

4.6.3. KAS FFC kdfKc

Contains properties **REQUIRED** for "kdfKc" registration.

Table 8 — kdfKc Capabilities

JSON Value	Description	JSON type	Valid Values	Optional
kdfOption	The kdf options supported	object	Section 4.10	No
dkmNonceTypes	The dkmNonceTypes supported	array of string	randomNonce, timestamp, sequence, timestampSequence	Required for dhStatic scheme
kcOption	The kc options supported	object	Section 4.11	No
parameterSet	The parameter sets supported	object	Section 4.7.1	No

4.7. Parameter Sets

4.7.1. KAS FFC Parameter Set

Each parameter set advertised is a self-contained JSON object using the following values. Note that at least one parameter set ("fb", "fc") **MUST** be provided.

Table 9 — KAS FFC Parameter Set Capabilities JSON Values

JSON Value	Description	JSON type	Valid Values	Optional
fb	The fb parameter set	object	See Section 4.7.2	Yes
fc	The fc parameter set	object	See Section 4.7.2	Yes

4.7.2. KAS FFC Parameter Set Details

fb/fc changes minimum allowed values on options.

- fb: Len p — 2048, Len q — 224, min hash len — 112, min keySize — 112, min macSize — 64
- fc: Len p — 2048, Len q — 256, min hash len — 112, min keySize — 128, min macSize — 64

"noKdfNoKc" **REQUIRES** "hashAlg"

"kdfNoKc" **REQUIRES** "hashAlg" and at least one valid MAC registration

"kdfKc" **REQUIRES** "hashAlg" and at least one valid MAC registration

Table 10 — KAS FFC Parameter Set Details Capabilities JSON Values

JSON Value	Description	JSON type	Valid Values	Optional
hashAlg	The hash algorithms to use for DSA (and noKdfNoKc)	array	See Section 4.8	Yes
macOption	The macOption(s) to use with "kdfNoKc" and/or "kdfKc"	object	See Section 4.9	Yes

4.8. Supported Hash Algorithm Methods

The following SHA methods **MAY** be advertised by the ACVP compliant crypto module:

- SHA-1
- SHA2-224
- SHA2-256
- SHA2-384
- SHA2-512

4.9. Supported KAS FFC MAC Options

The following MAC options **MAY** be advertised for registration under a “kdfNoKc” and “kdfKc” kasMode:

- AES-CCM
- CMAC
- HMAC-SHA-1
- HMAC-SHA2-224
- HMAC-SHA2-256
- HMAC-SHA2-384
- HMAC-SHA2-512

Table 11 — KAS FFC Mac Option Details

JSON Value	Description	JSON type	Valid Values	Optional
keyLen	The supported keyLens for the selected MAC.	Domain	AES based MACs limited to 128, 192, 256. HashAlg based MACs mod 8. All keySizes minimum **MUST** conform to parameter set requirements See Section 4.7.2 .	No
nonceLen	The nonce len for use with AES-CCM mac	value	Input as bits, 56-104, odd byte values only (7-13). Additionally minimum **MUST** conform to parameter set requirements See Section 4.7.2 .	Yes (**REQUIRED** for AES-CCM)
macLen	The mac len for use with mac	value	Input as bits, mod 8, minimum **MUST** conform to parameter set requirements See Section 4.7.2 , maximum **MAY NOT** exceed block size..	No

4.10. Supported KAS FFC KDF Options

The following MAC options are available for registration under a “kdfNoKc” and “kdfKc” kasMode:

- concatenation

Table 12 — KAS FFC KDF Option Details

JSON Value	Description	JSON type	Valid Values	Optional
oiPattern	The OI pattern to use for constructing OtherInformation.	value	See Section 4.10.1 .	No

4.10.1. Other Information Construction

Some IUTs **MAY** require a specific pattern for the OtherInfo portion of the KDFs for KAS. An “oiPattern” is specified in the KDF registration to accommodate such requirements. Regardless of the oiPattern specified, the OI bitlength **MUST** be 240 for FFC, and 376 for ECC. The OI will be padded with random bits (or the most significant bits utilized) when the specified OI pattern does not meet the bitlength requirement

Pattern candidates:

- literal[123456789ABCDEF]
  - uses the specified hex within “[]”. literal[123456789ABCDEF] substitutes “123456789ABCDEF” in place of the field
- uPartyInfo
  - uPartyId { || ephemeralKey } { || ephemeralNonce } { || dkmNonce }
    - dkmNonce is provided by party u for static schemes
    - “optional” items such as ephemeralKey **MUST** be included when available for ACVP testing.
- vPartyInfo { || ephemeralKey } { || ephemeralNonce }
  - vPartyId
    - “optional” items such as ephemeralKey **MUST** be included when available for ACVP testing.
- counter
  - 32bit counter starting at “1” (0×00000001)

Example (Note that party U is the server in this case “434156536964”, party V is the IUT “a1b2c3d4e5”, using an FFC non-static scheme):

- “concatenation” : “literal[123456789CAFECAFE]||uPartyInfo||vPartyInfo”

Evaluated as:

- “123456789CAFECAFE434156536964a1b2c3d4e5b16c5f78ef56e8c14a561”
  - “b16c5f78ef56e8c14a561” are random bits applied to meet length requirements

4.11. Supported KAS FFC KC Options

The following KC options are available for registration under a “kdfKc” kasMode:

Table 13 — KAS FFC KC Option Details Capabilities

JSON Value	Description	JSON type	Valid Values	Optional
kcRole	The role(s) the IUT is to act as for KeyConfirmation.	array	provider/recipient	No
kcType	The type(s) the IUT is to act as for KeyConfirmation.	array	unilateral/bilateral	No
nonceType	The nonce type(s) the IUT is to use for KeyConfirmation.	array	randomNonce, timestamp, sequence, timestampSequence	No

4.12. Example KAS FFC Capabilities JSON Object

The following is a example JSON object advertising support for KAS FFC.

```
{
 "algorithm": "KAS-FFC",
 "revision": "1.0",
 "prereqVals": [{
   "algorithm": "DSA",
   "valValue": "123456"
  },
  {
   "algorithm": "DRBG",
   "valValue": "123456"
  },
  {
   "algorithm": "SHA",
   "valValue": "123456"
  },
  {
   "algorithm": "CCM",
   "valValue": "123456"
  },
  {
   "algorithm": "CMAC",
   "valValue": "123456"
  },
  {
   "algorithm": "HMAC",
   "valValue": "123456"
  }
 ],
 "function": ["keyPairGen", "dpGen"],
```

```
"scheme": {
 "dhEphem": {
  "kasRole": ["initiator", "responder"],
  "kdfNoKc": {
   "kdfOption": {
    "concatenation": "uPartyInfo||vPartyInfo",
    "ASN1": "uPartyInfo||vPartyInfo"
   },
   "parameterSet": {
    "fc": {
     "hashAlg": ["SHA2-224", "SHA2-256"],
     "macOption": {
      "AES-CCM": {
       "keyLen": [128],
       "nonceLen": 56,
       "macLen": 64
      }
     }
    }
   }
  }
 },
 "mqv1": {
  "kasRole": ["initiator", "responder"],
  "kdfNoKc": {
   "kdfOption": {
    "concatenation": "uPartyInfo||vPartyInfo",
    "ASN1": "uPartyInfo||vPartyInfo"
   },
   "parameterSet": {
    "fc": {
     "hashAlg": ["SHA2-224", "SHA2-256"],
     "macOption": {
      "AES-CCM": {
       "keyLen": [128],
       "nonceLen": 56,
       "macLen": 64
      }
     }
    }
   }
  },
  "kdfKc": {
   "KcOption": {
    "KcRole": [
     "provider",
```

```
      "recipient"
     ],
     "KcType": [
      "unilateral",
      "bilateral"
     ],
     "NonceType": [
      "randomNonce"
     ]
    },
    "kdfOption": {
     "concatenation": "uPartyInfo||vPartyInfo",
     "ASN1": "uPartyInfo||vPartyInfo"
    },
    "parameterSet": {
     "fb": {
      "hashAlg": ["SHA2-224"],
      "macOption": {
       "AES-CCM": {
        "keyLen": [128],
        "nonceLen": 56,
        "macLen": 64
       }
      }
     },
     "fc": {
      "hashAlg": ["SHA2-256"],
      "macOption": {
       "AES-CCM": {
        "keyLen": [128],
        "nonceLen": 56,
        "macLen": 64
       }
      }
     }
    }
   }
  }
 }
}
```

Figure 2

4.13. Example KAS FFC Component Capabilities JSON Object

The following is a example JSON object advertising support for KAS FFC.

```
{
 "algorithm": "KAS-FFC",
 "mode": "Component",
 "revision": "1.0",
 "prereqVals": [{
   "algorithm": "DSA",
   "valValue": "123456"
  },
  {
   "algorithm": "DRBG",
   "valValue": "123456"
  },
  {
   "algorithm": "SHA",
   "valValue": "123456"
  },
  {
   "algorithm": "CCM",
   "valValue": "123456"
  },
  {
   "algorithm": "CMAC",
   "valValue": "123456"
  },
  {
   "algorithm": "HMAC",
   "valValue": "123456"
  }
 ],
 "function": ["keyPairGen", "dpGen"],
 "scheme": {
  "dhEphem": {
   "kasRole": ["initiator", "responder"],
   "noKdfNoKc": {
    "parameterSet": {
     "fb": {
      "hashAlg": ["SHA2-224", "SHA2-256"]
     }
    }
   }
  },
  "mqv1": {
   "kasRole": ["initiator", "responder"],
   "noKdfNoKc": {
    "parameterSet": {
     "fb": {
```

```
      "hashAlg": ["SHA2-224", "SHA2-256"]
     }
    }
   }
  }
 }
}
```

Figure 3

5. Generation requirements per party per scheme

The various schemes of KAS all have their own requirements as to keys and nonces per scheme, per party. The below table demonstrates those generation requirements:

Table 14 — Required Party Generation Obligations

Scheme	KasMode	KasRole	KeyConfirmationRo	KeyConfirmationDirecti	StaticKeyPa	EphemeralKeyP	EphemeralNo	PkmNon
DhHybrid1	NoKdfNoKc	InitiatorPartyU	None	None	True	True	False	False
DhHybrid1	NoKdfNoKc	ResponderPartyV	None	None	True	True	False	False
DhHybrid1	KdfNoKc	InitiatorPartyU	None	None	True	True	False	False
DhHybrid1	KdfNoKc	ResponderPartyV	None	None	True	True	False	False
DhHybrid1	KdfKc	InitiatorPartyU	Provider	Unilateral	True	True	False	False
DhHybrid1	KdfKc	InitiatorPartyU	Provider	Bilateral	True	True	False	False
DhHybrid1	KdfKc	InitiatorPartyU	Recipient	Unilateral	True	True	False	False
DhHybrid1	KdfKc	InitiatorPartyU	Recipient	Bilateral	True	True	False	False
DhHybrid1	KdfKc	ResponderPartyV	Provider	Unilateral	True	True	False	False
DhHybrid1	KdfKc	ResponderPartyV	Provider	Bilateral	True	True	False	False
DhHybrid1	KdfKc	ResponderPartyV	Recipient	Unilateral	True	True	False	False
DhHybrid1	KdfKc	ResponderPartyV	Recipient	Bilateral	True	True	False	False
Mqv2	NoKdfNoKc	InitiatorPartyU	None	None	True	True	False	False
Mqv2	NoKdfNoKc	ResponderPartyV	None	None	True	True	False	False
Mqv2	KdfNoKc	InitiatorPartyU	None	None	True	True	False	False
Mqv2	KdfNoKc	ResponderPartyV	None	None	True	True	False	False
Mqv2	KdfKc	InitiatorPartyU	Provider	Unilateral	True	True	False	False
Mqv2	KdfKc	InitiatorPartyU	Provider	Bilateral	True	True	False	False
Mqv2	KdfKc	InitiatorPartyU	Recipient	Unilateral	True	True	False	False
Mqv2	KdfKc	InitiatorPartyU	Recipient	Bilateral	True	True	False	False
Mqv2	KdfKc	ResponderPartyV	Provider	Unilateral	True	True	False	False
Mqv2	KdfKc	ResponderPartyV	Provider	Bilateral	True	True	False	False
Mqv2	KdfKc	ResponderPartyV	Recipient	Unilateral	True	True	False	False
Mqv2	KdfKc	ResponderPartyV	Recipient	Bilateral	True	True	False	False
DhEphem	NoKdfNoKc	InitiatorPartyU	None	None	False	True	False	False
DhEphem	NoKdfNoKc	ResponderPartyV	None	None	False	True	False	False
DhEphem	KdfNoKc	InitiatorPartyU	None	None	False	True	False	False
DhEphem	KdfNoKc	ResponderPartyV	None	None	False	True	False	False
DhHybridOneFlow	NoKdfNoKc	InitiatorPartyU	None	None	True	True	False	False
DhHybridOneFlow	NoKdfNoKc	ResponderPartyV	None	None	True	False	False	False
DhHybridOneFlow	KdfNoKc	InitiatorPartyU	None	None	True	True	False	False
DhHybridOneFlow	KdfNoKc	ResponderPartyV	None	None	True	False	False	False
DhHybridOneFlow	KdfKc	InitiatorPartyU	Provider	Unilateral	True	True	False	False
DhHybridOneFlow	KdfKc	InitiatorPartyU	Provider	Bilateral	True	True	False	False
DhHybridOneFlow	KdfKc	InitiatorPartyU	Recipient	Unilateral	True	True	False	False
DhHybridOneFlow	KdfKc	InitiatorPartyU	Recipient	Bilateral	True	True	False	False

Scheme	KasMode	KasRole	KeyConfirmationRo	KeyConfirmationDirecti	StaticKeyPa	EphemeralKeyP	EphemeralNo	OtkmNon
DhHybridOneFlow	KdfKc	ResponderPartyV	PartyV	Unilateral	True	False	False	False
DhHybridOneFlow	KdfKc	ResponderPartyV	PartyV	Bilateral	True	False	True	False
DhHybridOneFlow	KdfKc	ResponderPartyV	Recipient	Unilateral	True	False	True	False
DhHybridOneFlow	KdfKc	ResponderPartyV	Recipient	Bilateral	True	False	True	False
Mqv1	NoKdfNoKc	InitiatorPartyU	None	None	True	True	False	False
Mqv1	NoKdfNoKc	ResponderPartyV	None	None	True	False	False	False
Mqv1	KdfNoKc	InitiatorPartyU	None	None	True	True	False	False
Mqv1	KdfNoKc	ResponderPartyV	None	None	True	False	False	False
Mqv1	KdfKc	InitiatorPartyU	Provider	Unilateral	True	True	False	False
Mqv1	KdfKc	InitiatorPartyU	Provider	Bilateral	True	True	False	False
Mqv1	KdfKc	InitiatorPartyU	Recipient	Unilateral	True	True	False	False
Mqv1	KdfKc	InitiatorPartyU	Recipient	Bilateral	True	True	False	False
Mqv1	KdfKc	ResponderPartyV	PartyV	Unilateral	True	False	False	False
Mqv1	KdfKc	ResponderPartyV	PartyV	Bilateral	True	False	True	False
Mqv1	KdfKc	ResponderPartyV	Recipient	Unilateral	True	False	True	False
Mqv1	KdfKc	ResponderPartyV	Recipient	Bilateral	True	False	True	False
DhOneFlow	NoKdfNoKc	InitiatorPartyU	None	None	False	True	False	False
DhOneFlow	NoKdfNoKc	ResponderPartyV	None	None	True	False	False	False
DhOneFlow	KdfNoKc	InitiatorPartyU	None	None	False	True	False	False
DhOneFlow	KdfNoKc	ResponderPartyV	None	None	True	False	False	False
DhOneFlow	KdfKc	InitiatorPartyU	Recipient	Unilateral	False	True	False	False
DhOneFlow	KdfKc	ResponderPartyV	PartyV	Unilateral	True	False	False	False
DhStatic	NoKdfNoKc	InitiatorPartyU	None	None	True	False	False	False
DhStatic	NoKdfNoKc	ResponderPartyV	None	None	True	False	False	False
DhStatic	KdfNoKc	InitiatorPartyU	None	None	True	False	False	True
DhStatic	KdfNoKc	ResponderPartyV	None	None	True	False	False	False
DhStatic	KdfKc	InitiatorPartyU	Provider	Unilateral	True	False	False	True
DhStatic	KdfKc	InitiatorPartyU	Provider	Bilateral	True	False	False	True
DhStatic	KdfKc	InitiatorPartyU	Recipient	Unilateral	True	False	False	True
DhStatic	KdfKc	InitiatorPartyU	Recipient	Bilateral	True	False	False	True
DhStatic	KdfKc	ResponderPartyV	PartyV	Unilateral	True	False	False	False
DhStatic	KdfKc	ResponderPartyV	PartyV	Bilateral	True	False	True	False
DhStatic	KdfKc	ResponderPartyV	Recipient	Unilateral	True	False	True	False
DhStatic	KdfKc	ResponderPartyV	Recipient	Bilateral	True	False	True	False

6. Test Vectors

The ACVP server provides test vectors to the ACVP client, which are then processed and returned to the ACVP server for validation. A typical ACVP validation test session would require multiple test vector sets to be downloaded and processed by the ACVP client. Each test vector set represents an individual algorithm defined during the capability exchange. This section describes the JSON schema for a test vector set used with SP800-56a KAS FFC algorithms.

The test vector set JSON schema is a multi-level hierarchy that contains meta data for the entire vector set as well as individual test vectors to be processed by the ACVP client. The following table describes the JSON elements at the top level of the hierarchy.

Table 15 — Top Level Test Vector JSON Elements

JSON Values	Description	JSON Type
acvVersion	Protocol version identifier	string
vsId	Unique numeric vector set identifier	integer
algorithm	Algorithm defined in the capability exchange	string
mode	Mode defined in the capability exchange	string
revision	Protocol test revision selected	string
testGroups	Array of test groups containing test data, see Section 6.1	array

An example of this would look like this

```
{
  "acvVersion": "version",
  "vsId": 1,
  "algorithm": "Alg1",
  "mode": "Mode1",
  "revision": "Revision1.0",
  "testGroups": [ ... ]
}
```

Figure 4

6.1. Test Groups JSON Schema

The testGroups element at the top level in the test vector JSON object is an array of test groups. Test vectors are grouped into similar test cases to reduce the amount of data transmitted in the vector set. For instance, all test vectors that use the same key size MAY be grouped together. The Test Group JSON object contains meta data that applies to all test vectors within the group. The following table describes the secure hash JSON elements of the Test Group JSON object.

The test group for KAS FFC is as follows:

Table 16 — Vector Group JSON Object

JSON Value	Description	JSON type	Optional
tgId	Numeric identifier for the test group, unique across the entire vector set.	value	No
scheme	The scheme for the test vectors. See Section 4.5.1 for possible values	value	No
testType	The type of testCases expected within the group. AFT (Functional) tests produce test cases where the prompt file delivers only the needed public server information in which the IUT is expected to perform KAS. VAL (Validity) tests produce inputs/outputs from both server and IUT perspectives of a KAS negotiation. The expectation of the IUT on such tests is to determine if the KAS negotiation was successful or not.	AFT, VAL	No
kasRole	The KAS role	initiator, responder	No
kasMode	The KAS mode	noKdfNoKc, kdfNoKc, kdfKc	No
parmSet	Parameter set value to use	fb, fc	No
hashAlg	hashAlg values being used	See Section 4.8	No
macType	The MAC being used. **REQUIRED** for “kdfNoKc” and “kdfKc” modes.	See Section 4.9	Yes
keyLen	The key length of the MAC. **REQUIRED** for “kdfNoKc” and “kdfKc” modes.	See Section 4.9	Yes
nonceAesCcmLen	The nonce length of the MAC (applies only to AES-CCM). **REQUIRED** for	See Section 4.9	Yes

JSON Value	Description	JSON type	Optional
	"kdfNoKc" and "kdfKc" modes using a AES-CCM MAC.		
macLen	The mac length. **REQUIRED** for "kdfNoKc" and "kdfKc" modes.	See Section 4.9	Yes
kdfType	The KDF being used. **REQUIRED** for "kdfNoKc" and "kdfKc" modes.	concatenation, asn1	Yes
idServerLen	The length of the server ID. **REQUIRED** for "kdfNoKc" and "kdfKc" modes.	value	Yes
idServer	The server ID. **REQUIRED** for "kdfNoKc" and "kdfKc" modes.	value	Yes
idIutLen	The length of the server ID. **REQUIRED** for "kdfNoKc" and "kdfKc" modes. Provided in response by IUT for AFT tests.	value	Yes
idIut	The server ID. **REQUIRED** for "kdfNoKc" and "kdfKc" modes. Provided in response by IUT for AFT tests.	value	Yes
oiPattern	The oiPattern used in the KDF. For "kdfNoKc" and "kdfKc" modes.	See Section 4.10.1	Yes
kcRole	Key confirmation roles supported. **REQUIRED** for "kdfKc" modes.	provider, recipient	Yes
kcType	Key confirmation types supported. **REQUIRED** for "kdfKc" modes.	unilateral and/or bilateral	Yes
	Domain parameter for DSA	value	No
	Domain parameter for DSA	value	No
	Domain parameter for DSA	value	No
tests	Array of individual test vector JSON objects,	array	No

JSON Value	Description	JSON type	Optional
	which are defined in Section 6.2		

6.2. Test Case JSON Schema

Each test group contains an array of one or more test cases. Each test case is a JSON object that represents a single test vector to be processed by the ACVP client. The following table describes the JSON elements for each KAS FFC test vector.

Table 17 — Test Case JSON Object

JSON Value	Description	JSON type	Optional
tcId	Numeric identifier for the test case, unique across the entire vector set.	value	No
staticPublicServer	The DSA static public key	value	Yes
ephemeralPublicServer	The DSA ephemeral public key	value	Yes
nonceEphemeralServer	nonceEphemeralServer ONLY USED BY C(1,2) and C(0,2) schemes with KC. nonce to be used in the MacData field	value	Yes
nonceNoKc	The 16 byte nonce concatenated to the “Standard Test Message”. **REQUIRED** for No Key Confirmation tests only.	value	Yes
nonceDkm	The nonce supplied by the initiator to be used in the OI field in the PartyUInfo field.	value	Yes
staticPrivateIut	The IUT DSA static private key	value	Yes
staticPublicIut	The IUT DSA static public key	value	Yes
ephemeralPrivateIut	The IUT DSA ephemeral private key	value	Yes
ephemeralPublicIut	The IUT DSA ephemeral public key	value	Yes
oiLen	Length of the OtherInfo field	value	Yes
oi	OtherInfo field	value	Yes
dkm	Derived Keying Material.	value	Yes
tagIut	The tag (or MAC) GENERATED BY THE SERVER/IUT by using the DKM to MAC the Message with the specified method	value	Yes
nonceEphemeralIut	nonceEphemeralIut ONLY USED BY C(1,2) and C(0,2) schemes with KC. nonce to be used in the MacData field	value	Yes

JSON Value	Description	JSON type	Optional
nonceDkmIut	ONLY USED BY STATIC SCHEME. The nonce supplied by the initiator to be used in the OI field in the PartyUInfo field	value	Yes
nonceLenDkm	ONLY USED BY STATIC SCHEME. The length of the nonce supplied by the initiator to be used in the OI field in the PartyUInfo field.	value	Yes
nonceEphemeralDkm	ONLY USED BY C(1,2) and C(0,2) schemes with KC. nonce to be used in the MacData field	value	Yes
nonceEphemralDkmLen	length of nonceEphemeralIut value.	value	Yes
nonceAesCcm	Nonce used by the CCM function, if CCM is used to generate the Tag.	value	Yes
macData	The message to be MAced.	value	Yes
	A shared secret that is used to derive secret keying material using a key derivation function.	value	Yes
hashZServer	The hashed shared secret, only provided in noKdfNoKc modes of operation.	value	Yes
hashZIut	The hashed shared secret, only provided in noKdfNoKc modes of operation.	value	Yes
testPassed	Pass Fail indicating if the IUT agrees with the Tag generated by the server.	boolean	Yes

6.3. Example Test Vectors JSON Object

The following is a example JSON object for KAS FFC test vectors sent from the ACVP server to the crypto module.

```
[{
  "acvVersion": "version"
 },
 {
  "vsId": 1564,
  "algorithm": "KAS-FFC",
  "revision": "1.0",
  "testGroups": [
   {
```

```
    "tgId": 1,
    "scheme": "dhEphem",
    "testType": "AFT",
    "kasRole": "responder",
    "kasMode": "kdfNoKc",
    "parmSet": "fc",
    "hashAlg": "SHA2-226",
    "macType": "AES-CCM",
    "keyLen": 128,
    "aesCcmNonceLen": 56,
    "macLen": 64,
    "kdfType": "asn1",
    "idServerLen": 48,
    "idServer": "434156536964",
    "oiPattern": "uPartyInfo||vPartyInfo",
    "p":
"91d80f6b6958b04e87da7e3258db8dd60ba376cdfabdd635ce04a69e62bb34202f2a1f8fc4460a12cd2f265a
    "q": "cf4ff0acc36d5d9573342664b38b9a363173cfec110ecebc2dc3ef74780ca313",
    "g":
"15a498872f9758485c0253357b957e67ffbf320d955fec5dcf8d17acf715bb2de1de7872efb4c5902fc71c5c
    "tests": [{
     "tcId": 4000,
     "ephemeralPublicServer":
"4def03e88563f4992120c7940dff3b2c6da81c82a4b132c52916fd48811f0d6ce96fe7224e477df3a594566d
     "nonceNoKc": "83fffe9a2cb8f97868c2e5d140ce427a"
    }]
   },
   {
    "tgId": 2,
    "scheme": "dhEphem",
    "testType": "VAL",
    "kasRole": "initiator",
    "kasMode": "kdfNoKc",
    "parmSet": "fc",
    "hashAlg": "SHA2-226",
    "macType": "AES-CCM",
    "keyLen": 128,
    "aesCcmNonceLen": 56,
    "macLen": 64,
    "kdfType": "asn1",
    "idServerLen": 48,
    "idServer": "434156536964",
    "idIutLen": 40,
    "idIut": "a1b2c3d4e5",
    "oiPattern": "uPartyInfo||vPartyInfo",
```

```
    "p":
"d5cf9ba288ff8438650904a5fe2eeb8bf6b52b691a455b21bd3b37998f82544036ae61fe436039c66feab83b
    "q": "ceb9916bbc14cdc9dda80481135bee68ee94f4ecadc2921261a316d1c9cf9283",
    "g":
"37a7b2a094ed6253784d51c71a4a7407f0727df15e480a02b0ff2f9befdb0e92c0d63e482c6909e5337373ab
    "tests": [{
     "tcId": 5000,
     "ephemeralPublicServer":
"4fc5ed02abf0791bb01ad33527bd16ee58f875bbc85de5b513ccce95d05d6c735e726a102f11a69152f1c430
     "nonceNoKc": "1d33ef74a9f3c6e7ab1dfa77b40e4191",
     "ephemeralPrivateIut":
"41f8582a2f3ac19cc925fb5cf72ec98f89630fec8e71853397870c3fda1eb08b",
     "ephemeralPublicIut":
"a6f3b734391112b2ae81cb3021bc1275383c4a6c80b9a1265ae3f5a8256a0a621667f71e3e081b86bc427d5a
     "oiLen": 240,
     "oi": "a1b2c3d4e54341565369646cfd9fa9ec70ae7f9b0d17cc63ea2103fbaf6b",
     "nonceAesCcm": "6526d522a19f25",
     "tagIut": "9a0ade2f4b22599e"
    }]
   },
   {
    "tgId": 3,
    "scheme": "mqv1",
    "testType": "AFT",
    "kasRole": "initiator",
    "kasMode": "kdfNoKc",
    "parmSet": "fb",
    "hashAlg": "SHA2-512",
    "macType": "HMAC-SHA2-256",
    "keyLen": 128,
    "macLen": 128,
    "kdfType": "asn1",
    "idServerLen": 48,
    "idServer": "434156536964",
    "p":
"E4E85DB7DF0C5AA669C26DFA7A8920C0580975FA36F761347C081F9631C120C1EFA7E1165859B1564F7EFA2E
    "q": "A1B3E7100F528EE0824F2A5A6F3B022F1ADFBFF02307F5FCC0224F5D",
    "g":
"1AB425960434544374CA03EDAF1AEA9ADC3EACF233D2FDE6A594A58C1BB93F08C8D01B7CE7160FD439A2E642
    "tests": [{
     "tcId": 421,
     "staticPublicServer":
"6E71B7F2E5690D279348B24B06B4341B77389942C3A6497CA9A3E7066BF84CF9DE750E235FE14EC80E17361A
     "nonceNoKc": "1ED327DA0EF730995949D0DFAA00ADCA"
    }]
   },
```

```
  {
   "tgId": 4,
   "scheme": "mqv1",
   "testType": "AFT",
   "kasRole": "responder",
   "kasMode": "kdfNoKc",
   "parmSet": "fb",
   "hashAlg": "SHA2-512",
   "macType": "HMAC-SHA2-384",
   "keyLen": 128,
   "macLen": 128,
   "kdfType": "asn1",
   "idServerLen": 48,
   "idServer": "434156536964",
   "p":
"E4E85DB7DF0C5AA669C26DFA7A8920C0580975FA36F761347C081F9631C120C1EFA7E1165859B1564F7EFA2E
   "q": "A1B3E7100F528EE0824F2A5A6F3B022F1ADFBFF02307F5FCC0224F5D",
   "g":
"1AB425960434544374CA03EDAF1AEA9ADC3EACF233D2FDE6A594A58C1BB93F08C8D01B7CE7160FD439A2E642
   "tests": [{
    "tcId": 441,
    "staticPublicServer":
"37EA4A6FE4FB2D86BAB256B987DECFEAAADA9C26CA74D39EEBEE71341BD09DE22C124EC1679AE1604F0F1BFE
    "ephemeralPublicServer":
"A109397DC12D1EC800C9E8617C2291DFCAD91221ADAA444816B3ADC825AF86C3DCCCF24EBBB1AE8AF9C47E0E
    "nonceNoKc": "DC04E3B3C2E0C0F14CDB425D04B48E58"
   }]
  },
  {
   "tgId": 5,
   "scheme": "mqv1",
   "testType": "VAL",
   "kasRole": "initiator",
   "kasMode": "kdfNoKc",
   "parmSet": "fb",
   "hashAlg": "SHA2-512",
   "macType": "CMAC",
   "keyLen": 256,
   "macLen": 128,
   "kdfType": "asn1",
   "idServerLen": 48,
   "idServer": "434156536964",
   "idIutLen": 0,
   "p":
"E4E85DB7DF0C5AA669C26DFA7A8920C0580975FA36F761347C081F9631C120C1EFA7E1165859B1564F7EFA2E
   "q": "A1B3E7100F528EE0824F2A5A6F3B022F1ADFBFF02307F5FCC0224F5D",
```

```
    "g":
"1AB425960434544374CA03EDAF1AEA9ADC3EACF233D2FDE6A594A58C1BB93F08C8D01B7CE7160FD439A2E642
    "tests": [{
     "tcId": 461,
     "staticPublicServer":
"9959B0B4D8E61B80642D654FF8029E93FA05111D4001CFE2867FBCBCF6084202CA99BB7B6E7245C731F23436
     "nonceNoKc": "DB4614836A5B33947EA8D38268026E81",
     "staticPrivateIut":
"11B2F386A6ACBEBA7D09F504B4D52BDF9F0E0F72097FDE2DCD65E6A4",
     "staticPublicIut":
"3771A4584C3A30238D8F1173EE34C6BAEF78F37C30D1A2159AA9147DC530FCADDB950198C1473063C4D9AC2F
     "ephemeralPrivateIut":
"5610BA8A79D92592D44BF65A634E7CA81BA276F4023CBE01F888B7D7",
     "ephemeralPublicIut":
"3DD2FB97DDD2A59B209146806CEF7218BF53445FB5F6ED4444E608A923BDEE4BA84C78B0ED1A50513B77985F
     "oiLen": 240,
     "oi": "A1B2C3D4E5434156536964CAFECAFE0C702E54817AD2878111B540EE479C",
     "tagIut": "72E1D9346A2840777B3860D9B930A411"
    }]
   },
   {
    "tgId": 6,
    "scheme": "mqv1",
    "testType": "VAL",
    "kasRole": "responder",
    "kasMode": "kdfNoKc",
    "parmSet": "fb",
    "hashAlg": "SHA2-512",
    "macType": "HMAC-SHA2-256",
    "keyLen": 128,
    "macLen": 128,
    "kdfType": "asn1",
    "idServerLen": 48,
    "idServer": "434156536964",
    "idIutLen": 0,
    "p":
"E4E85DB7DF0C5AA669C26DFA7A8920C0580975FA36F761347C081F9631C120C1EFA7E1165859B1564F7EFA2E
    "q": "A1B3E7100F528EE0824F2A5A6F3B022F1ADFBFF02307F5FCC0224F5D",
    "g":
"1AB425960434544374CA03EDAF1AEA9ADC3EACF233D2FDE6A594A58C1BB93F08C8D01B7CE7160FD439A2E642
    "tests": [{
     "tcId": 511,
     "staticPublicServer":
"5009DB149D378FCD02BADFEA1AD5A4A1FAFA90E6144F10211D325E977C9FFC38457AA85E0885E35551EA5EAF
     "ephemeralPublicServer":
"29C3A5172136C0D17DE1C7B46C7B8E300E49C00E485DBDB0F2349F912B1D649C5A0EDF69D196B3061AF90FF2
```

```
    "nonceNoKc": "6BC3341BD73345D9771F842A44768699",
    "staticPrivateIut":
"31DFEBA03A0F42B3CA14F45B413527C0CB78AB9FC840F6781F3F1546",
    "staticPublicIut":
"33CE93E95F8C7C8CAC844D24CA346AD3287B0AC983222EA5012F49A88776A6FB177F0020C029464D42D0B448
    "oiLen": 240,
    "oi": "434156536964A1B2C3D4E5CAFECAFE71CAD14BEB7B210AD4BAA4D9CDC8E1",
    "tagIut": "F20532B36666B0204DE17195E9C4DC89"
   }]
  },
  {
   "tgId": 7,
   "scheme": "mqv1",
   "testType": "AFT",
   "kasRole": "initiator",
   "kasMode": "kdfKc",
   "parmSet": "fb",
   "hashAlg": "SHA2-224",
   "macType": "AES-CCM",
   "keyLen": 256,
   "aesCcmNonceLen": 64,
   "macLen": 128,
   "kdfType": "asn1",
   "idServerLen": 48,
   "idServer": "434156536964",
   "kcRole": "provider",
   "kcType": "unilateral",
   "p":
"DAEA7C71588BFCA3B8ED40E103CE87E49701D228822CF4CD3A5A5FBA7F07895EEA4928234724E50521054CCD
   "q": "F3C65FD74CBA905970D1E32925D7FD88E2412AA88C1614E579890A75",
   "g":
"61376016993BBA8A477E30C8A78C5846F02A6A8D7A80664C26FA815CFA7243A528400F86013960B1E78D9EE9
   "tests": [{
    "tcId": 561,
    "staticPublicServer":
"82122FBA230C65ECAFE305ED20FA3C640E15344C3093D3E3BE2907F16D4EABD6CD4876E3D7C136BFF31DC83D
    "nonceEphemeralServer":
"3D8C2739CE915A3E2E67B252679200E15240B659C57539E4639C4F354CBB9736D8A51C2DD522CE0597B1998D
    "nonceAesCcm": "37CC8408E6C20632"
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  {
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   "testType": "AFT",
   "kasRole": "initiator",
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    "kasMode": "kdfKc",
    "parmSet": "fb",
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    "macType": "AES-CCM",
    "keyLen": 256,
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    "q": "F3C65FD74CBA905970D1E32925D7FD88E2412AA88C1614E579890A75",
    "g":
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    "tests": [{
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     "nonceEphemeralServer":
"0F3B5A93A6DFE6EE5C422F48CEF718A199ECF411DED531D83C12525D5AAADAEC26114BD6F41DB81146B3890E
     "nonceAesCcm": "E660DC439C9112B0"
    }]
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    "testType": "AFT",
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    "parmSet": "fb",
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    "keyLen": 128,
    "macLen": 128,
    "kdfType": "asn1",
    "idServerLen": 48,
    "idServer": "434156536964",
    "kcRole": "recipient",
    "kcType": "unilateral",
    "p":
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    }]
   },
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    "testType": "AFT",
    "kasRole": "initiator",
    "kasMode": "kdfKc",
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    "macType": "AES-CCM",
    "keyLen": 256,
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    "kdfType": "asn1",
    "idServerLen": 48,
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    "kcRole": "recipient",
    "kcType": "bilateral",
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     "nonceEphemeralServer":
"CA85E0A9365BFB31E9276C69169B28EEA553BAA32F386C4582FE5CCC80F4EB4A3890B09B0F8A8DE47FA3AD96
     "nonceAesCcm": "2F186CD57426963A"
    }]
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    "testType": "AFT",
    "kasRole": "responder",
    "kasMode": "kdfKc",
    "parmSet": "fb",
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    "macType": "AES-CCM",
    "keyLen": 256,
    "aesCcmNonceLen": 64,
    "macLen": 128,
    "kdfType": "asn1",
    "idServerLen": 48,
    "idServer": "434156536964",
    "kcRole": "provider",
    "kcType": "unilateral",
    "p":
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    "g":
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     "ephemeralPublicServer":
"0F911BCDE004699FD4FCC863A37BE5776BA36C9C1F05FB5A0439E9B0668E5AB9346EC9F4DF3DE455C0D96F65
     "nonceAesCcm": "D0D5B47C412EEF48"
    }]
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    "testType": "AFT",
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    "kasMode": "kdfKc",
    "parmSet": "fb",
    "hashAlg": "SHA2-224",
    "macType": "AES-CCM",
    "keyLen": 256,
    "aesCcmNonceLen": 64,
    "macLen": 128,
    "kdfType": "asn1",
    "idServerLen": 48,
    "idServer": "434156536964",
    "kcRole": "provider",
    "kcType": "bilateral",
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    "q": "F3C65FD74CBA905970D1E32925D7FD88E2412AA88C1614E579890A75",
    "g":
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     "ephemeralPublicServer":
"3AD77DACE248A4A680EC9EE722EBA556BC156F0AB49B14319313791D71438FD89880C1E57C502A2D82C7B513
     "nonceAesCcm": "EDE971BC462678E6"
    }]
   },
   {
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    "scheme": "mqv1",
    "testType": "AFT",
    "kasRole": "responder",
    "kasMode": "kdfKc",
    "parmSet": "fb",
    "hashAlg": "SHA2-224",
    "macType": "AES-CCM",
    "keyLen": 256,
    "aesCcmNonceLen": 64,
    "macLen": 128,
    "kdfType": "asn1",
    "idServerLen": 48,
    "idServer": "434156536964",
    "kcRole": "recipient",
    "kcType": "unilateral",
    "p":
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    "q": "F3C65FD74CBA905970D1E32925D7FD88E2412AA88C1614E579890A75",
    "g":
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     "ephemeralPublicServer":
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     "nonceAesCcm": "35DEA4B3B6FE3427"
    }]
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   {
    "tgId": 14,
    "scheme": "mqv1",
    "testType": "AFT",
    "kasRole": "responder",
    "kasMode": "kdfKc",
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    "parmSet": "fb",
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    "kcRole": "recipient",
    "kcType": "bilateral",
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    "q": "F3C65FD74CBA905970D1E32925D7FD88E2412AA88C1614E579890A75",
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     "ephemeralPublicServer":
"5C6719CEE3A9002AC457A78CE8214E24B23F06970EED6DBB44AF4410C4AEB9B3A777CCB103187F089E0391B3
     "nonceAesCcm": "B45755807B729C8D"
    }]
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    "testType": "VAL",
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    "kasMode": "kdfKc",
    "parmSet": "fb",
    "hashAlg": "SHA2-224",
    "macType": "AES-CCM",
    "keyLen": 256,
    "aesCcmNonceLen": 64,
    "macLen": 128,
    "kdfType": "asn1",
    "idServerLen": 48,
    "idServer": "434156536964",
    "idIutLen": 0,
    "kcRole": "provider",
    "kcType": "unilateral",
    "p":
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    "q": "F3C65FD74CBA905970D1E32925D7FD88E2412AA88C1614E579890A75",
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"1E9B67AECABEC61FBF7666205E1EFE0F1656A241C1F4106EB4411497E1723C1210AF9DA000A2BFC053033E84
     "staticPrivateIut":
"E3FCC5CEFCE5FE934289C79622486E9CD80D14489E95519A4A15A10A",
     "staticPublicIut":
"CD25A49630B46B2261D2B3123C2B3D6C73E26785E4F1CD2667BA20C5FCE3BDC05562D5BD0751F81100C0E218
     "ephemeralPrivateIut":
"33D1DD6C1F7161F55DECE654F5CD5728421FEE931A78B31326589B2A",
     "ephemeralPublicIut":
"86F7404F39265D26599824759D421D09DAE89C8BF71B229B993D2F82A902A3559D3F0A0B3B5EFD74CB218988
     "oiLen": 240,
     "oi": "A1B2C3D4E5434156536964CAFE12346DE2A120592CBF724D26FD3854648D",
     "nonceAesCcm": "D19A63EEA4CBD552",
     "tagIut": "675E9100D4D2437C16697FA7078AB354"
    }]
   },
   {
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    "scheme": "mqv1",
    "testType": "VAL",
    "kasRole": "initiator",
    "kasMode": "kdfKc",
    "parmSet": "fb",
    "hashAlg": "SHA2-224",
    "macType": "AES-CCM",
    "keyLen": 256,
    "aesCcmNonceLen": 64,
    "macLen": 128,
    "kdfType": "asn1",
    "idServerLen": 48,
    "idServer": "434156536964",
    "idIutLen": 0,
    "kcRole": "provider",
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    "nonceEphemeralServer":
"E6FC63D94A63C77E55FECE73FDEE073DC82D9D30C0A65E75EA76E7D2894C0E95DB58B222C467868E6666D8B1
    "staticPrivateIut":
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    "staticPublicIut":
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    "ephemeralPrivateIut":
"CB3F1D45D40AC9D89FB3A4F70DC9DFFAD19438E0E016A650C47E68B5",
    "ephemeralPublicIut":
"D51C6F1214D83C2D0695DA2E85CE153848B8FF511220DB4ADA7831B67171AC4C4D70952B8E59F7A24554851C
    "oiLen": 240,
    "oi": "A1B2C3D4E5434156536964CAFE12344F588C8270ABDEB66A5FC6170247D0",
    "nonceAesCcm": "9F697AC59FAFAC14",
    "tagIut": "DD8630597AC383B79424268E560966A5"
   }]
  },
  {
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   "scheme": "mqv1",
   "testType": "VAL",
   "kasRole": "initiator",
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   "hashAlg": "SHA2-224",
   "macType": "AES-CCM",
   "keyLen": 256,
   "aesCcmNonceLen": 64,
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   "idServerLen": 48,
   "idServer": "434156536964",
   "idIutLen": 0,
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   "kcType": "unilateral",
   "p":
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    "staticPublicIut":
"16CDF697CC07832AB9BBDECB5D18F3C7F2840B1286407FF18E8AB1133643A9D8A37BBAC357886D4D1B953153
    "ephemeralPrivateIut":
"89AE5C57C633F93FCEC3CA468C03C9C915848EBF0D854D8D7F0A0748",
    "ephemeralPublicIut":
"4FABB7AC021C510FBB092E1C0070BA16FC71AEFE5D2283C127672A15D0E7B9223BA455EFB4B9D03E6FF4ED0A
    "oiLen": 240,
    "oi": "A1B2C3D4E5434156536964CAFE123459D3C3FDDEEE446BD96F4FFA30EF41",
    "nonceAesCcm": "57D68AFA30D5994D",
    "tagIut": "83F4FD5FB96722E8FE4386C04C0BDC3C"
   }]
  },
  {
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   "kasMode": "kdfKc",
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   "macType": "AES-CCM",
   "keyLen": 256,
   "aesCcmNonceLen": 64,
   "macLen": 128,
   "kdfType": "asn1",
   "idServerLen": 48,
   "idServer": "434156536964",
   "idIutLen": 0,
   "kcRole": "recipient",
   "kcType": "bilateral",
   "p":
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   "q": "F3C65FD74CBA905970D1E32925D7FD88E2412AA88C1614E579890A75",
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"A30DC780FF140C17EF497261C2C8F5E54117CBEF63B89A17820D1B6997CF0B84FA344C84D7CA27EBC3B2B16C
    "staticPrivateIut":
"86586780FED8E66478E55CEB38AE4EE085EF4C87ACB542A3DF343D6C",
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    "staticPublicIut":
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    "ephemeralPrivateIut":
"0EB4D2A40A2E49E5AAFC792BEE2183368CEF504DA813305B55318ACA",
    "ephemeralPublicIut":
"10B29FA275824C2305BD414494330ADC8FA284E27D75840979573C210A1975685A842BC19EEEE0F462A6A6F6
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    "oi": "A1B2C3D4E5434156536964CAFE1234A490B28BFA803CE2ED934F0AE8DCCA",
    "nonceAesCcm": "B682D26D3E3857A8",
    "tagIut": "25CAEE5040F7AA018D82C5066AD7881F"
   }]
  },
  {
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   "scheme": "mqv1",
   "testType": "VAL",
   "kasRole": "responder",
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   "macType": "AES-CCM",
   "keyLen": 256,
   "aesCcmNonceLen": 64,
   "macLen": 128,
   "kdfType": "asn1",
   "idServerLen": 48,
   "idServer": "434156536964",
   "idIutLen": 0,
   "kcRole": "provider",
   "kcType": "unilateral",
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   "q": "F3C65FD74CBA905970D1E32925D7FD88E2412AA88C1614E579890A75",
   "g":
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    "ephemeralPublicServer":
"B3F76590E308FEB6F4C7B0C0AE68EB250A400F94C06AC56DB020D6745B7F6385D3206C36C75016EBE3751A24
    "staticPrivateIut":
"278FB569309A162DDA6E41F24BCBB2D514AC393C80F3E0CB32843FE7",
    "staticPublicIut":
"2EC4CBC9E993CDD608D904198F9B7F487F6FEE261B37BDDF45FD17CBAD4CF9D4057F042D3FE33ADEE9F8C402
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    "tagIut": "266167B5B251C379479EF80F282B637C"
   }]
  },
  {
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   "macType": "AES-CCM",
   "keyLen": 256,
   "aesCcmNonceLen": 64,
   "macLen": 128,
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   "idServerLen": 48,
   "idServer": "434156536964",
   "idIutLen": 0,
   "kcRole": "provider",
   "kcType": "bilateral",
   "p":
"DAEA7C71588BFCA3B8ED40E103CE87E49701D228822CF4CD3A5A5FBA7F07895EEA4928234724E50521054CCD
   "q": "F3C65FD74CBA905970D1E32925D7FD88E2412AA88C1614E579890A75",
   "g":
"61376016993BBA8A477E30C8A78C5846F02A6A8D7A80664C26FA815CFA7243A528400F86013960B1E78D9EE9
   "tests": [{
    "tcId": 1791,
    "staticPublicServer":
"689ADE017EB260D8DBE1A160557FCF260729CE5FCB3094520C62EC49F94CCA74CDE6CA19053D6C0FA6A98750
    "ephemeralPublicServer":
"387A751726FC8EACD5804E11DD249C72D56E460A4D1C5D6FBD22B9BC0BCF034C10B2E1896539F547F0D997C8
    "staticPrivateIut":
"01964A4D06931493AB48CE3AAEC7B6F9214EF8E68FAD285F67F9C6C8",
    "staticPublicIut":
"B45CD5C273435636EC9AE3FE94295BC2D8B483CEB41E4B933CCE4A90972F4F862819F8EFF2FD638C83A084C8
    "nonceEphemeralIut":
"799A4D6D4FA5C842F03E7CE9845FEAE7A1E8A9C55A49664407C3014610CBDACCBA64BE824B6CE280AC88AF7D
    "oiLen": 240,
    "oi": "434156536964A1B2C3D4E5CAFE1234F90CE7E9F0B4A48D1986BCCA62B35D",
    "nonceAesCcm": "925048AAB27B4358",
    "tagIut": "E8CA0A03AAA7DC5C8A4F2ECB7E9DDB64"
   }]
  },
```

```
  {
   "tgId": 21,
   "scheme": "mqv1",
   "testType": "VAL",
   "kasRole": "responder",
   "kasMode": "kdfKc",
   "parmSet": "fb",
   "hashAlg": "SHA2-224",
   "macType": "AES-CCM",
   "keyLen": 256,
   "aesCcmNonceLen": 64,
   "macLen": 128,
   "kdfType": "asn1",
   "idServerLen": 48,
   "idServer": "434156536964",
   "idIutLen": 0,
   "kcRole": "recipient",
   "kcType": "unilateral",
   "p":
"DAEA7C71588BFCA3B8ED40E103CE87E49701D228822CF4CD3A5A5FBA7F07895EEA4928234724E50521054CCD
   "q": "F3C65FD74CBA905970D1E32925D7FD88E2412AA88C1614E579890A75",
   "g":
"61376016993BBA8A477E30C8A78C5846F02A6A8D7A80664C26FA815CFA7243A528400F86013960B1E78D9EE9
   "tests": [{
    "tcId": 1941,
    "staticPublicServer":
"D91479786D5E975DBF77B6B2425AB0AC68C593E291DF9CB8B56A52A8BA4469E583ACEFB62430E24633A573B2
    "ephemeralPublicServer":
"03019E8B2EF4688CD771C9A1B032E09541DBCFB4AC9EDA3B9C965D175DD45E4F3642CB31694EAD9331FA0597
    "staticPrivateIut":
"D97952EE78B1C2431AC50868E756AC14C6B29D18883515D1665D023A",
    "staticPublicIut":
"D0E1766A6B479411B50AD54961E042FEF80429DFA17D863381D4853EF080700D2B4DC262040EC4592DDBA5A9
    "nonceEphemeralIut":
"E5D188DAD2A9E863287AEBA62D059921BC13DFE9451344FD767E5C5DA5E78957AB24E0F7E67580C98179D7DF
    "oiLen": 240,
    "oi": "434156536964A1B2C3D4E5CAFE123498CB185732655116484B422B2B9B6E",
    "nonceAesCcm": "507A464107DB9D05",
    "tagIut": "7906E07684F1B1DFC75CD8EA6BC36EFF"
   }]
  },
  {
   "tgId": 22,
   "scheme": "mqv1",
   "testType": "VAL",
   "kasRole": "responder",
```

```
    "kasMode": "kdfKc",
    "parmSet": "fb",
    "hashAlg": "SHA2-224",
    "macType": "AES-CCM",
    "keyLen": 256,
    "aesCcmNonceLen": 64,
    "macLen": 128,
    "kdfType": "asn1",
    "idServerLen": 48,
    "idServer": "434156536964",
    "idIutLen": 0,
    "kcRole": "recipient",
    "kcType": "bilateral",
    "p":
"DAEA7C71588BFCA3B8ED40E103CE87E49701D228822CF4CD3A5A5FBA7F07895EEA4928234724E50521054CCD
    "q": "F3C65FD74CBA905970D1E32925D7FD88E2412AA88C1614E579890A75",
    "g":
"61376016993BBA8A477E30C8A78C5846F02A6A8D7A80664C26FA815CFA7243A528400F86013960B1E78D9EE9
    "tests": [{
     "tcId": 2091,
     "staticPublicServer":
"96A947FFF446538F7315DA2C0DCD341FC459A63D4FB8D253CA4120C3B8406B60F4BE0BED9EFA038C27314CEC
     "ephemeralPublicServer":
"15A5F4C5B439293DD59E031531346DA8C776046F9E4F058B6A4B9E927E1F1B50CA62B2FA46E68C4488BECF83
     "staticPrivateIut":
"A905C2AD3791CA57115E0175EBB1846F657B7725990CA575EB648CFC",
     "staticPublicIut":
"3C98E5319B90663E31B4C0E3163DAA6F0F0792A1610908D35DEDD785890B1962EC89BE8C83EF60DF50435CD6
     "nonceEphemeralIut":
"40B82C42063C4D4002C655FB65F36A38BD3B4B6713256FFE5DAFB1BC2D452E26858194F9B27B9FE1E1FC8D89
     "oiLen": 240,
     "oi": "434156536964A1B2C3D4E5CAFE12346069DA63735D4E63DD752238101F06",
     "nonceAesCcm": "055D20623BA47053",
     "tagIut": "DBB039CE12C4D4D6E1C89B84028FD875"
    }]
   }
  ]
 }
]
```


Figure 5

6.4. Example Test Vectors Component JSON Object

The following is a example JSON object for KAS FFC test vectors sent from the ACVP server to the crypto module.

```
[{
  "acvVersion": "version"
 },
 {
  "vsId": 1564,
  "algorithm": "KAS-FFC",
  "mode": "Component",
  "revision": "1.0",
  "testGroups": [{
    "tgId": 1,
    "scheme": "dhEphem",
    "testType": "AFT",
    "kasRole": "initiator",
    "kasMode": "noKdfNoKc",
    "parmSet": "fb",
    "hashAlg": "SHA2-512",
    "p":
"a3a77cce3b0ea9891fe6ac34b2bdd04e22f9fd5a4976b5e2bd4c9ec43831c4d293779f3c4f826e6c2a8d6bd1
    "q": "f094f4fa8fa36fdcdf4f0378112bfde03cfa532e666b9736b5ab76e9",
    "g":
"45308211a07f231181276b44b873eb67726ca6aa5ecd39b4274f780409e15bfc98ac4680be5220a23b963e3b
    "tests": [{
     "tcId": 3000,
     "ephemeralPublicServer":
"2cc5dcb3cae0bd0052838991e371fca0bb827598c1ffa554f2699ccc29b5bb085bf77634ab3fec24fb81ab1e
    }]
   },

   {
    "tgId": 2,
    "scheme": "dhEphem",
    "testType": "VAL",
    "kasRole": "responder",
    "kasMode": "noKdfNoKc",
    "parmSet": "fb",
    "hashAlg": "SHA2-512",
    "p":
"b0b4760d008b63403f2ce5ae597d0adfa56a11da5c29301b25565282c0fda5fc634d51fd738807800b506e47
    "q": "947d217b8f4d658f0f9c6d427d28dcc2bb18b186f0754aac47c9cae5",
    "g":
"a7ae223bb4f3360505170ce0524015523d212b1746a24ecf78eca58e572227b92cfea96c6769b57ddd74e0dc
    "tests": [{
     "tcId": 6000,
     "ephemeralPublicServer":
"4fc5ed02abf0791bb01ad33527bd16ee58f875bbc85de5b513ccce95d05d6c735e726a102f11a69152f1c430
```

```
    "ephemeralPrivateIut":
"1DB3456FFA3A40B37D2EFD7B51084340E101280A48DC412BA4C6FC89",
    "ephemeralPublicIut":
"4A712F5AA0F4E72BD12E07F708EF937468615D865051847C9BA19751A4CA92F3071F42D2646C55EA773530F4
    "hashZIut":
"6a8a50b45c32dc4e0eca0447822024e843dbbad378392ca4d2da83ad40500fe2dd06f5fab677725e5706dd20
   }]
  },
  {
   "tgId": 3,
   "scheme": "mqv1",
   "testType": "AFT",
   "kasRole": "initiator",
   "kasMode": "noKdfNoKc",
   "parmSet": "fb",
   "hashAlg": "SHA2-512",
   "p":
"E4E85DB7DF0C5AA669C26DFA7A8920C0580975FA36F761347C081F9631C120C1EFA7E1165859B1564F7EFA2E
   "q": "A1B3E7100F528EE0824F2A5A6F3B022F1ADFBFF02307F5FCC0224F5D",
   "g":
"1AB425960434544374CA03EDAF1AEA9ADC3EACF233D2FDE6A594A58C1BB93F08C8D01B7CE7160FD439A2E642
   "tests": [{
    "tcId": 281,
    "staticPublicServer":
"58573EBC7A52B7FC467A512EA4D40E5262F4E66C48742B5433A2AB0D040796F6E718865B22149BCBB25E17F1
   }]
  },
  {
   "tgId": 4,
   "scheme": "mqv1",
   "testType": "AFT",
   "kasRole": "responder",
   "kasMode": "noKdfNoKc",
   "parmSet": "fb",
   "hashAlg": "SHA2-512",
   "p":
"E4E85DB7DF0C5AA669C26DFA7A8920C0580975FA36F761347C081F9631C120C1EFA7E1165859B1564F7EFA2E
   "q": "A1B3E7100F528EE0824F2A5A6F3B022F1ADFBFF02307F5FCC0224F5D",
   "g":
"1AB425960434544374CA03EDAF1AEA9ADC3EACF233D2FDE6A594A58C1BB93F08C8D01B7CE7160FD439A2E642
   "tests": [{
    "tcId": 301,
    "staticPublicServer":
"6C679621CD67F982CAC083BF7ED8938C80BB336A7CE5F752274E006EBA7F9F4F91F87414FEF3040711F6D0DC
    "ephemeralPublicServer":
"14A0B1C99306E9D152D33C9D0C36855B45A320EDB4C2EF93A3460403D42AF220BF3F835CB57503AE3D959793
```

```
    }]
   },
   {
    "tgId": 5,
    "scheme": "mqv1",
    "testType": "VAL",
    "kasRole": "initiator",
    "kasMode": "noKdfNoKc",
    "parmSet": "fb",
    "hashAlg": "SHA2-512",
    "p":
"E4E85DB7DF0C5AA669C26DFA7A8920C0580975FA36F761347C081F9631C120C1EFA7E1165859B1564F7EFA2E
    "q": "A1B3E7100F528EE0824F2A5A6F3B022F1ADFBFF02307F5FCC0224F5D",
    "g":
"1AB425960434544374CA03EDAF1AEA9ADC3EACF233D2FDE6A594A58C1BB93F08C8D01B7CE7160FD439A2E642
    "tests": [{
     "tcId": 321,
     "staticPublicServer":
"B853A15F818A3748CA8E160E2A12E3CF2D517E5002C29497A072164861F4A069C3D867999EEBF560633686B7
     "staticPrivateIut":
"03271FB1CB05B4C2A45A067CABB560B3BAF9550DA0E26299E0567874",
     "staticPublicIut":
"AD62D4942746A618B69078E411766A62C1355C5015ABB970326B01B080A1DCD5D6A006EACF9B72F2746BD053
     "ephemeralPrivateIut":
"1DB3456FFA3A40B37D2EFD7B51084340E101280A48DC412BA4C6FC89",
     "ephemeralPublicIut":
"4A712F5AA0F4E72BD12E07F708EF937468615D865051847C9BA19751A4CA92F3071F42D2646C55EA773530F4
     "hashZIut":
"408FB2DA722F83A8BEF6E4B11B61D6EE3F68F5063AF0F6FF767F9AEDC19454346220F8342D24B4FC1F3A950F
    }]
   },
   {
    "tgId": 6,
    "scheme": "mqv1",
    "testType": "VAL",
    "kasRole": "responder",
    "kasMode": "noKdfNoKc",
    "parmSet": "fb",
    "hashAlg": "SHA2-512",
    "p":
"E4E85DB7DF0C5AA669C26DFA7A8920C0580975FA36F761347C081F9631C120C1EFA7E1165859B1564F7EFA2E
    "q": "A1B3E7100F528EE0824F2A5A6F3B022F1ADFBFF02307F5FCC0224F5D",
    "g":
"1AB425960434544374CA03EDAF1AEA9ADC3EACF233D2FDE6A594A58C1BB93F08C8D01B7CE7160FD439A2E642
    "tests": [{
     "tcId": 371,
```

```
    "staticPublicServer":
 "99EA49651F5B300334549841A1076A0DDE62CBF8645393CD755AB265FC68D6A1DF82518E950013E4B0BB7A8B
    "ephemeralPublicServer":
 "922B5A91261A6B3891877BF72A82E8A0E2B2236F55A6FA46684488F5BE906D2EC61AC21799E69511E445786D
    "staticPrivateIut":
 "49E0A35472359ACD8A57678D6C2C8BA39CA51773686844FEA93231A8",
    "staticPublicIut":
 "2E07FCECF6689EF17EF2186194A7A065ACBEB75118397F48934647A42F3A95EF92695844E6091F8DCEE3D51C
    "hashZIut":
 "3AFFD88F2D1240C0AACBE43BE34178B69CC65E0215DCF9B126977769C5850F3E04551FC4CE47EC9ED219EA80
   }]
  }
 ]
}
]
```


Figure 6

7. Test Vector Responses

After the ACVP client downloads and processes a vector set, it **SHALL** send the response vectors back to the ACVP server. The following table describes the JSON object that represents a vector set response.

7.1. Vector Set Response JSON Object

Table 18 — Vector Set Response JSON Object

JSON Value	Description	JSON type
acvVersion	Protocol version identifier	value
vsId	Unique numeric identifier for the vector set	value
testGroups	Array of JSON objects that represent each test vector group. See Section 7.2	array

The testGroups section is used to organize the ACVP client response in a similar manner to how it receives vectors. Several algorithms SHALL require the client to send back group level properties in their response. This structure helps accommodate that.

7.2. Vector Set Group Response JSON Object

Table 19 — Vector Set Group Response JSON Object

JSON Value	Description	JSON type
tgId	The test group Id	value
tests	The tests associated to the group specified in tgId	value

7.3. Example Test Results JSON Object

The following is a example JSON object for KAS FFC test results sent from the crypto module to the ACVP server.

```
[{
  "acvVersion": "1.0"
 },
 {
  "vsId": 1564,
  "testGroups": [{
    "tgId": 1,
    "tests": [{
     "tcId": 4000,
     "nonceAesCcm": "67dc43e8d722e9",
     "ephemeralPublicIut":
"55b25962987f35576d371abe075741d4b20b04e989d4ea6dcee252b3d90aee720445816127ad49d4d8144e6f
     "iutIdLen": 40,
```

```
     "iutId": "a1b2c3d4e5",
     "oiLen": 240,
     "oi": "434156536964a1b2c3d4e52009a24c98c92d8a86461fbb212c8193db7a69",
     "dkm": "72d5339d984529cf25ecca527c273282",
     "tagIut": "fa3ab95a72aac9c0"
    }]
   },
   {
    "tgId": 2,
    "tests": [{
     "tcId": 5000,
     "testPassed": true
    }]
   },
   {
    "tgId": 3,
    "tests": [{
     "tcId": 421,
     "nonceNoKc": "1ED327DA0EF730995949D0DFAA00ADCA",
     "staticPublicIut":
"C915231FFD84E47A3584708102B28B4370EB8CB0EFC3E98F36B658D2ABFADD481D19D0B501553E7B095E1B66
     "ephemeralPublicIut":
"B9A09B0644960023F5E56E1CFCAA3EFC87AE7D5A3674053E8F16C7602BB49B4033774AA4C09F47635D3AB461
     "idIutLen": 40,
     "idIut": "A1B2C3D4E5",
     "oiLen": 240,
     "oi": "A1B2C3D4E5434156536964CAFECAFE3E3188BE5726763063BBC77815C4CC",
     "tagIut": "7F7A2CDFAA8B7C18E76D0FF243A7611A"
    }]
   },
   {
    "tgId": 4,
    "tests": [{
     "tcId": 441,
     "nonceNoKc": "DC04E3B3C2E0C0F14CDB425D04B48E58",
     "staticPublicIut":
"7B08F4AFA4D47B8CD75C7C8A777983153007268B09250648ACB476E7E61A6441FA2E9E2160A14D2CD452BA25
     "idIutLen": 40,
     "idIut": "A1B2C3D4E5",
     "oiLen": 240,
     "oi": "434156536964A1B2C3D4E5CAFECAFE6C9DC1203604F366F59FD2CD1D06BB",
     "tagIut": "69B9840FF2ECE380843658FF6D84AEBD"
    }]
   },
   {
    "tgId": 5,
```

```
    "tests": [{
     "tcId": 461,
     "testPassed": true
    }]
   },
   {
    "tgId": 6,
    "tests": [{
     "tcId": 511,
     "testPassed": true
    }]
   },
   {
    "tgId": 7,
    "tests": [{
     "tcId": 561,
     "staticPublicIut":
"10AA2F9E9574A42B0F99DBD5F1A56F13AA5B90CD2728F7E5A0EADE798C0CDE62B6FEAA55DB9CFA6BE92EB3EA
     "ephemeralPublicIut":
"815229B735BDC1AED41649920687737A7A913EC74CE050A17678B5550A549F2255B7A79C96B6A10ADE102A5C
     "idIutLen": 40,
     "idIut": "A1B2C3D4E5",
     "oiLen": 240,
     "oi": "A1B2C3D4E5434156536964CAFE1234DFD3219FB9293975E00069B19B786B",
     "nonceAesCcm": "37CC8408E6C20632",
     "tagIut": "DD6BDAABD4395A0DE058F7FE38C74E99"
    }]
   },
   {
    "tgId": 8,
    "tests": [{
     "tcId": 621,
     "staticPublicIut":
"79CB5FBE3926516FC6061570EF17B2D8B657FE21E60A15335EDBB7010902167C0A743C894595E1401317E151
     "ephemeralPublicIut":
"3E96095F6A9BD9869EF46BFF0DCD6FA2C998B1C6274D735DCC1FC06E152D4B2606CFA5F844E9DA240C37DAEA
     "idIutLen": 40,
     "idIut": "A1B2C3D4E5",
     "oiLen": 240,
     "oi": "A1B2C3D4E5434156536964CAFE12346485B01764B240882CA93582E428D0",
     "nonceAesCcm": "E660DC439C9112B0",
     "tagIut": "EE9B29AF0C99A4467473CBB62640887B"
    }]
   },
   {
    "tgId": 9,
```

```
    "tests": [{
     "tcId": 701,
     "staticPublicIut":
"5C9E65226A043001C097C1783534FD0D83CC6241E90D23BFF3F1DA6398C98351339947D5BF39E46028B66502
     "ephemeralPublicIut":
"96AC319C263ED125BC72A44A672DAE6AF5106206F06F71FC2B7410EFA611A39F4892F34ADB553B3E00AE6D2E
     "idIutLen": 40,
     "idIut": "A1B2C3D4E5",
     "oiLen": 240,
     "oi": "A1B2C3D4E5434156536964CAFE12340D4994262F58263CE2A09FB84E1936",
     "tagIut": "C89F779369214060CE5616A76C3C57C3"
    }]
   },
   {
    "tgId": 10,
    "tests": [{
     "tcId": 741,
     "staticPublicIut":
"4DAEDC3087655E4A1711063D04773B346D6182701263C83F2488E483D94A34250D3B97575CBB55FE0287ED11
     "ephemeralPublicIut":
"D814144AA499FA7A5F8CDB6FD643AFBF0719F46F7CA66B5647F5940616E3E448E8C80C394041BD79CEE869A5
     "idIutLen": 40,
     "idIut": "A1B2C3D4E5",
     "oiLen": 240,
     "oi": "A1B2C3D4E5434156536964CAFE12349D4938C5E98D90A0B7C87EAD5BB5A7",
     "nonceAesCcm": "2F186CD57426963A",
     "tagIut": "FBBA6F4C22B9576E33D50168A57BCA52"
    }]
   },
   {
    "tgId": 11,
    "tests": [{
     "tcId": 801,
     "staticPublicIut":
"C365B6B78380B5E84C1F467D30C1B5222BC84881D75C2255163727D4D1BEA003AB91AE226149DB71609F903A
     "idIutLen": 40,
     "idIut": "A1B2C3D4E5",
     "oiLen": 240,
     "oi": "434156536964A1B2C3D4E5CAFE12344BF1519741142E9F0FB37B10F6C15C",
     "nonceAesCcm": "D0D5B47C412EEF48",
     "tagIut": "3B8FC751CA4DB7EB379DDBFD7A254727"
    }]
   },
   {
    "tgId": 12,
    "tests": [{
```

```
      "tcId": 861,
      "staticPublicIut":
"BA4ADE415B3E0A2A88595BE522D77A2963A5B843C0F8E457861A9BDBE8BF4B8769BED1A0D822E988B1089A91
      "nonceEphemeralIut":
"DF3C33CC7A92236708D9CDEFDC25AD0BE92C13E03C1376710A27C922B1BFB5610474A64A801B81FC031B5F28
      "idIutLen": 40,
      "idIut": "A1B2C3D4E5",
      "oiLen": 240,
      "oi": "434156536964A1B2C3D4E5CAFE123432318DE92AF1C6652ABDDAFE55EEC0",
      "nonceAesCcm": "EDE971BC462678E6",
      "tagIut": "5EA618600A8286A912DE41B376117757"
     }]
    },
    {
     "tgId": 13,
     "tests": [{
      "tcId": 921,
      "staticPublicIut":
"A1A36FEF2BF595E6F121E5DDC06BB73E309A602BE7559111766CB89A4C0A525DEBCCC26E3A86130381DED487
      "nonceEphemeralIut":
"1D57658278BB2DDB8C284C7E5FCC875E2A1F5DDE68E5C2FB07373D1799EFD20438BCB654DAD4009D33C54914
      "idIutLen": 40,
      "idIut": "A1B2C3D4E5",
      "oiLen": 240,
      "oi": "434156536964A1B2C3D4E5CAFE123465F78C6B19AFBF62FE90B780163332",
      "nonceAesCcm": "35DEA4B3B6FE3427",
      "tagIut": "F8D33BFEB681D38E69EFB88C837093F4"
     }]
    },
    {
     "tgId": 14,
     "tests": [{
      "tcId": 981,
      "staticPublicIut":
"746BD74D66E42D7705FE779EE9B26620867E72CFC9B1759259395B17243800A52EC6B97D785BDDCAC62E95C0
      "nonceEphemeralIut":
"6EDF0CCE8BD793CB76F46769C90B2F7778A86DD16DA555DCBB102A0894887174A19178D752274A712276A863
      "idIutLen": 40,
      "idIut": "A1B2C3D4E5",
      "oiLen": 240,
      "oi": "434156536964A1B2C3D4E5CAFE1234F959F05CEA78C5C1F03F9BE78ABAA6",
      "nonceAesCcm": "B45755807B729C8D",
      "tagIut": "CC6DADBD739FE9ACCE2F1A29C710DF76"
     }]
    },
    {
```

```
    "tgId": 15,
    "tests": [{
     "tcId": 1041,
     "testPassed": true
    }]
   },
   {
    "tgId": 16,
    "tests": [{
     "tcId": 1191,
     "testPassed": true
    }]
   },
   {
    "tgId": 17,
    "tests": [{
     "tcId": 1341,
     "testPassed": true
    }]
   },
   {
    "tgId": 18,
    "tests": [{
     "tcId": 1491,
     "testPassed": true
    }]
   },
   {
    "tgId": 19,
    "tests": [{
     "tcId": 1641,
     "testPassed": false
    }]
   },
   {
    "tgId": 20,
    "tests": [{
     "tcId": 1791,
     "testPassed": false
    }]
   },
   {
    "tgId": 21,
    "tests": [{
     "tcId": 1941,
     "testPassed": true
```

```
    }]
   },
   {
    "tgId": 22,
    "tests": [{
     "tcId": 2091,
     "testPassed": false
    }]
   }
  ]
 }
]
```

Figure 7

7.4. Example Test Results Component JSON Object

The following is a example JSON object for KAS FFC Component test results sent from the crypto module to the ACVP server.

```
[{
  "acvVersion": "version"
 },
 {
  "vsId": 1564,
  "testGroups": [{
    "tgId": 1,
    "tests": [{
     "tcId": 3000,
     "ephemeralPublicIut":
 "5067495e06d0c6dfd6eada5607f3dc7b9bf93eeeb7513119860f5d60ef332823b0ba58af2ecb7ab2cbccee87
     "hashZIut":
 "39de343caa24b36f02ce0d3b08a7c5f553c7f5f0f65dcca10855a3fed89efd475156b51bfdfb1509e7898ca7
    }]
   },
   {
    "tgId": 2,
    "tests": [{
     "tcId": 5000,
     "testPassed": true
    }]
   }
  ]
 }
]
```

Figure 8

8. Security Considerations

There are no additional security considerations outside of those outlined in the ACVP document.

Appendix A — Terminology

For the purposes of this document, the following terms and definitions apply.

A.1.

Prompt
JSON sent from the server to the client describing the tests the client performs
Registration
The initial request from the client to the server describing the capabilities of one or several algorithm, mode and revision combinations
Response
JSON sent from the client to the server in response to the prompt
Test Case
An individual unit of work within a prompt or response
Test Group
A collection of test cases that share similar properties within a prompt or response
Test Vector Set
A collection of test groups under a specific algorithm, mode, and revision
Validation
JSON sent from the server to the client that specifies the correctness of the response

Appendix B — Abbreviations and Acronyms

ACVP	Automated Crypto Validation Protocol
JSON	Javascript Object Notation

Appendix C — Revision History

Table C-1

Version	Release Date	Updates
1	2018-09-01	Initial Release

Appendix D — References

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